

NB40: Bayes Error Ceiling - Tum Paneller

SEED=42 | k-NN + Label Tutarsizligi | 2026-06-25

1. Bayes Hatasi Nedir?

Bayes hatasi (R^*) bir siniflandirma probleminde mumkun olan en iyi modelin bile yapacagi minimum hatadir. Kaynagi model degil, verinin kendisindeki belirsizliktir. Ornek: iki varyant ayni feature degerleri tasiyor ama biri benign biri pathogenic ise, hic bir model bunlari ayiramaz. Bu hata indirgenemez (irreducible).

Bu notebook, k-NN Cover-Hart sinirlari + label tutarsizligi ölçutleri kullanarak her panelde Bayes hatasi bandini (R^*_{low} , R^*_{high}) tahmin eder, ve Bayes-optimal F1 tavanini son test dagilimlarina (final %80/20) gore hesaplar.

2. Sonuc Ozeti (Tum Paneller)

Panel	n	kNN-1	R^*_{low}	R^*_{high}	Bayes-F1	Model-F1	Gap
MASTER	2931	0.289	0.175	0.289	0.654	0.638	0.016
KANSER	388	0.198	0.112	0.198	0.726	0.730	-0.004
PAH	372	0.218	0.124	0.218	0.694	0.582	0.112
CFTR	111	0.189	0.106	0.189	0.755	0.863	-0.108

3. Karar Tablosu

Panel	Gap	Karar
MASTER	0.016	DUR
KANSER	-0.004	DUR
PAH	0.112	DEVAM
CFTR	-0.108	DUR

Karar Kurali: Gap < 0.03 = DUR | 0.03-0.10 = ARA | Gap > 0.10 = DEVAM

4. Temel Bulgular

CFTR: F1=0.755, gap=-0.108 (low conf)

KANSER: F1=0.726, gap=-0.004

MASTER: F1=0.654, gap=0.016

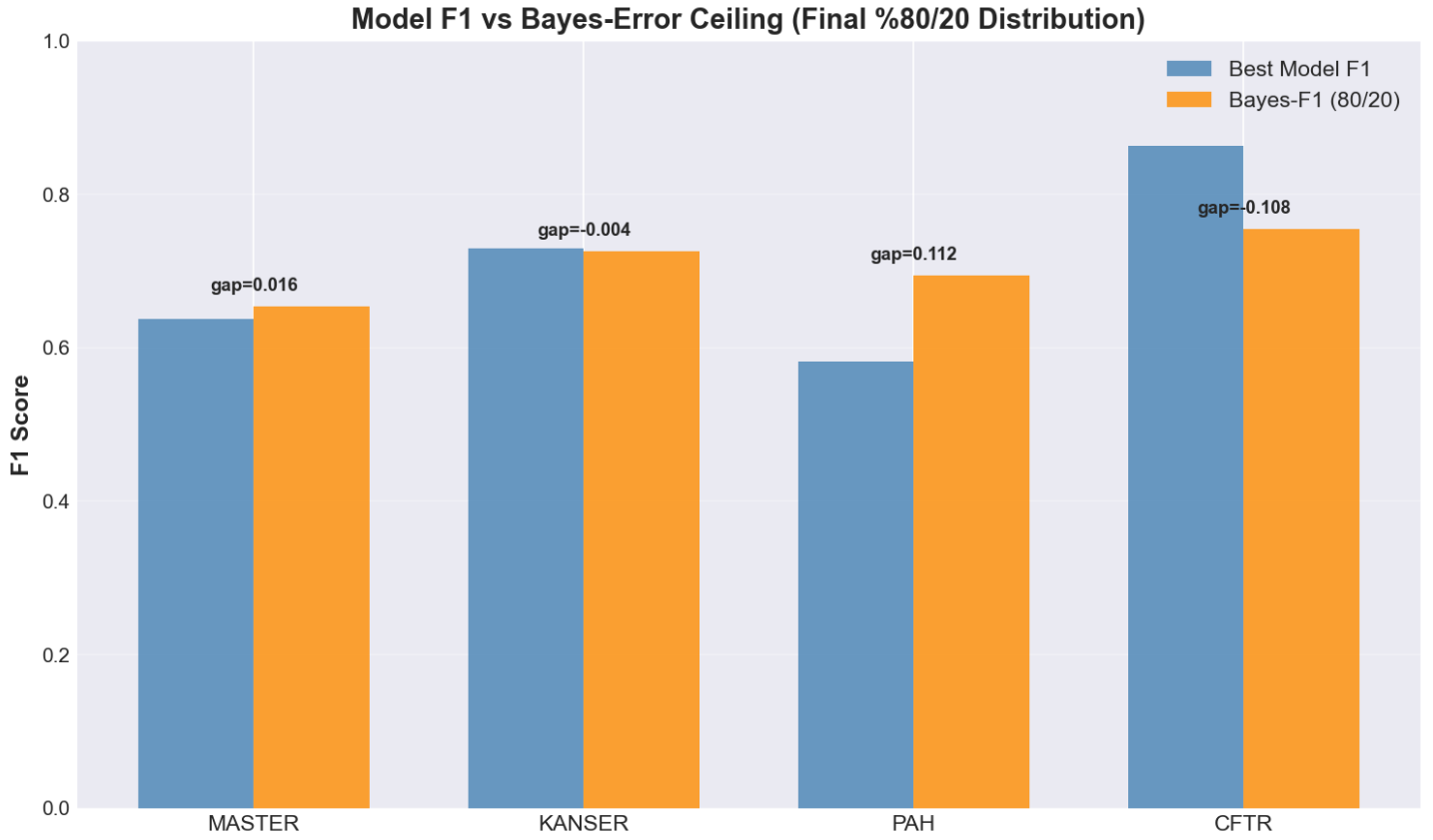
PAH: F1=0.694, gap=0.112

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5. Gorsellestirilmis Sonuclar

Sekil 1: fig1_model_vs_ceiling.png

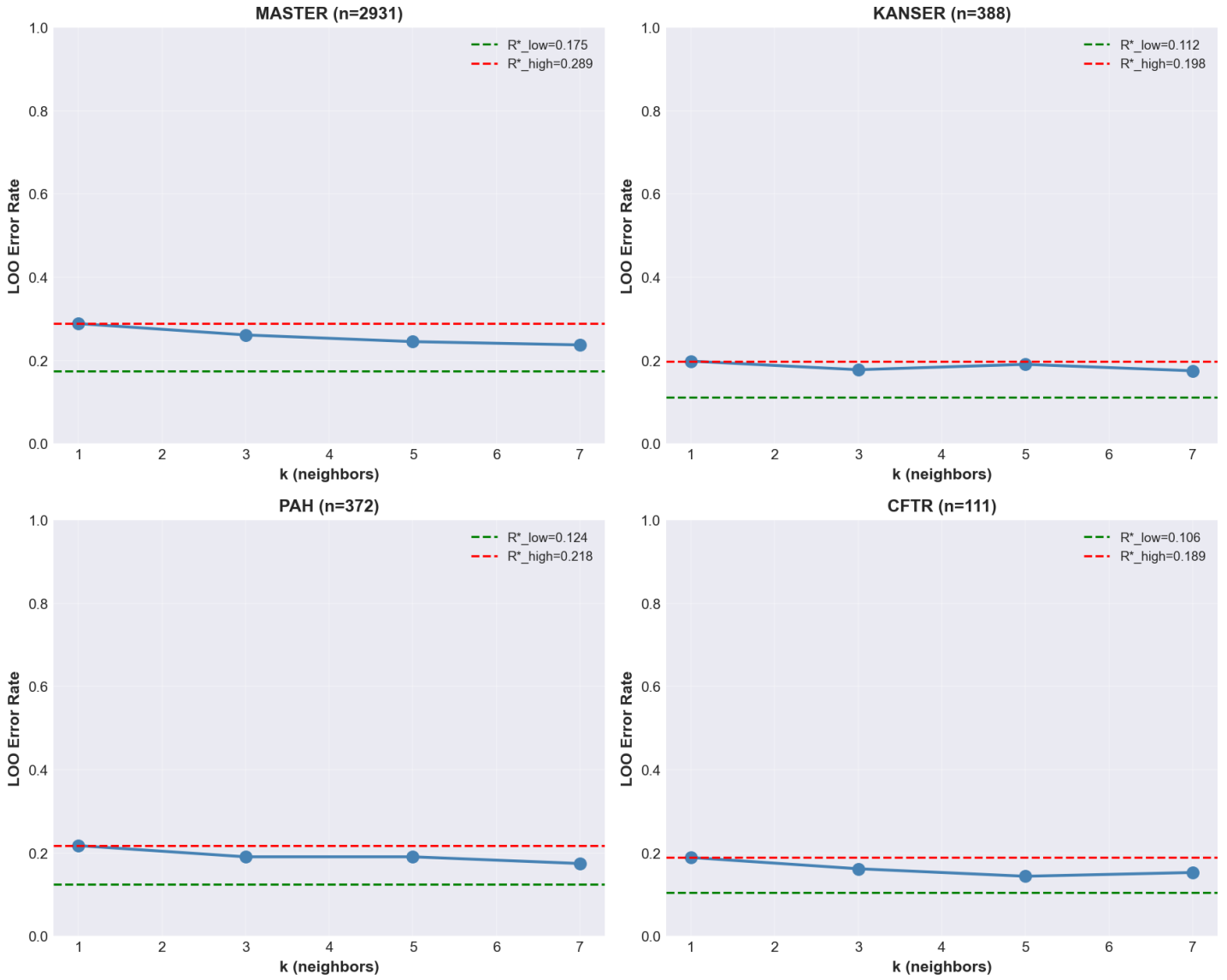


Sekil 2: fig2_knn_curve.png

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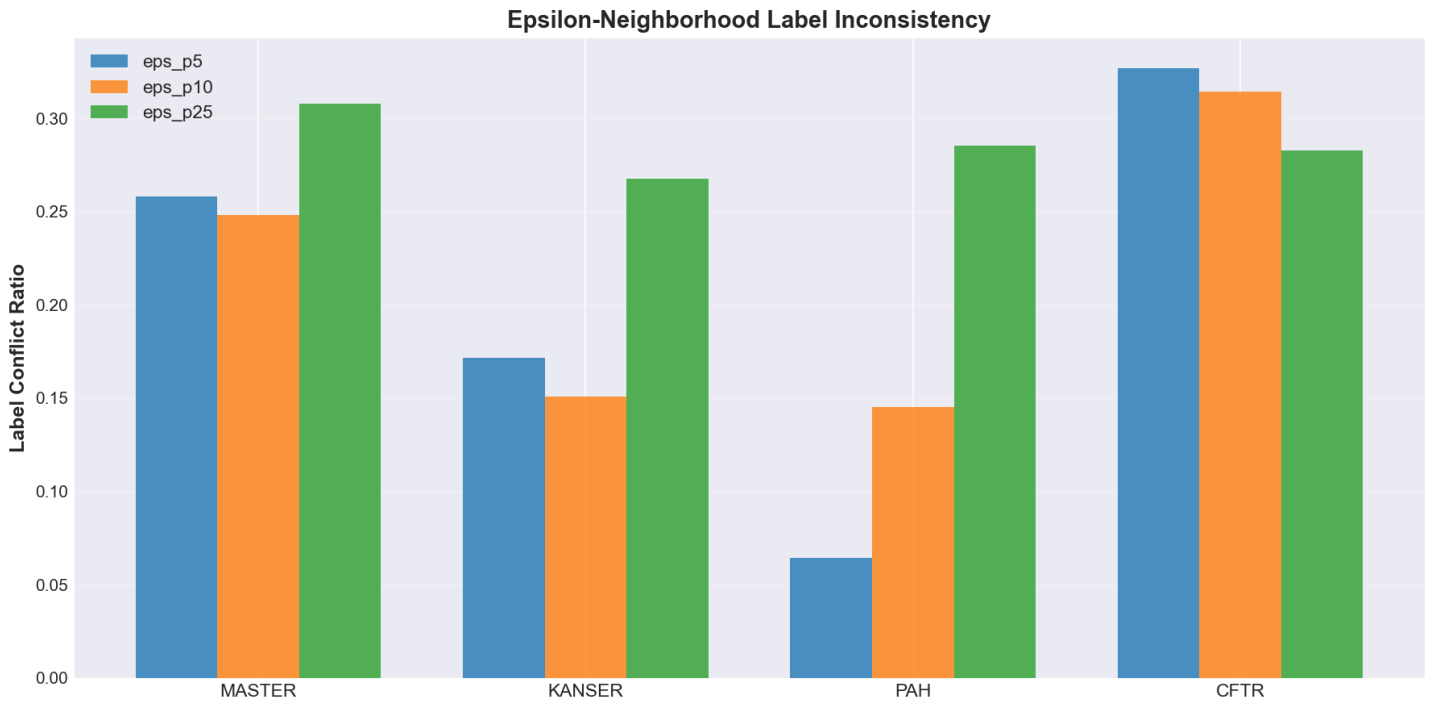
k-NN Leave-One-Out Error Rates & Cover-Hart Bounds



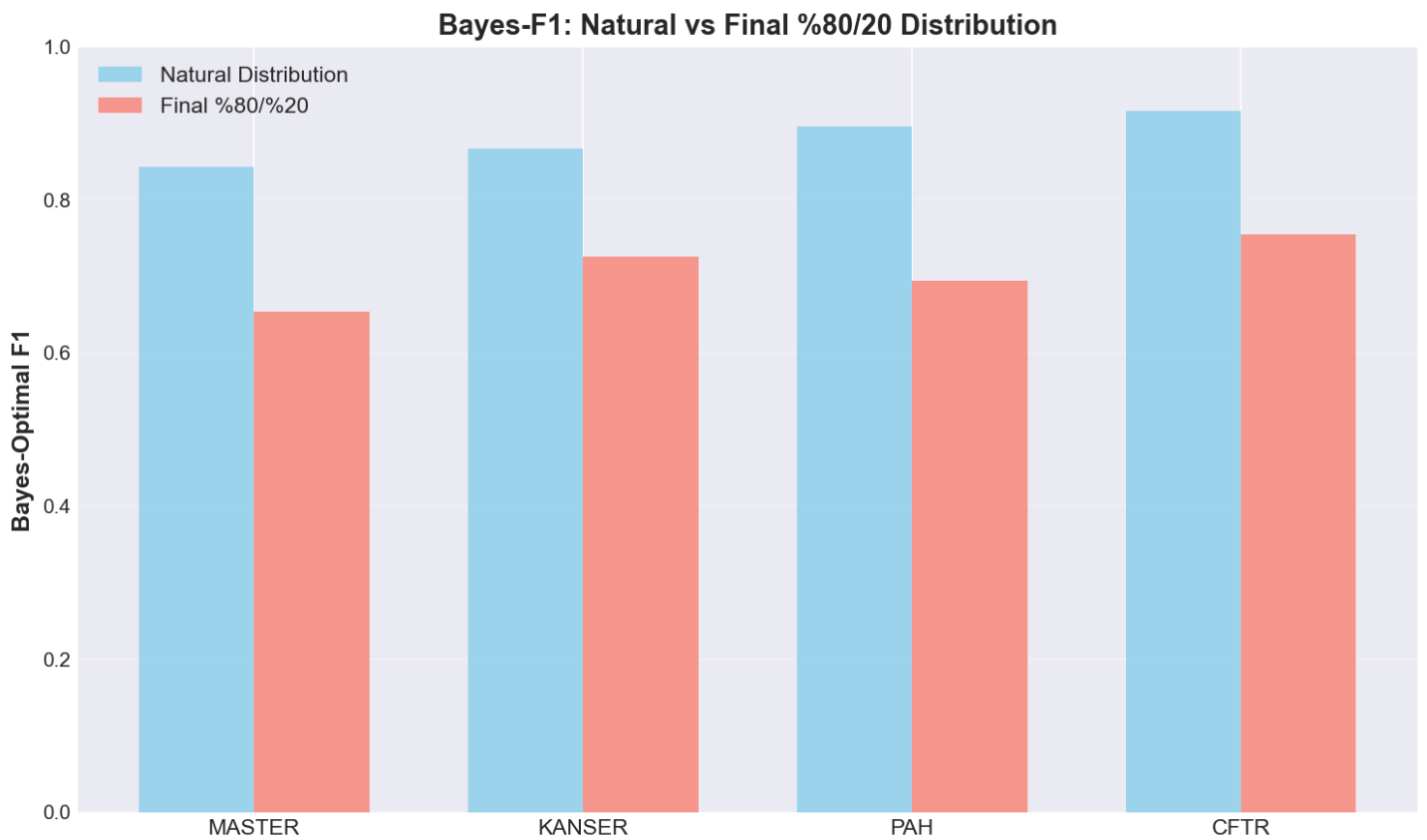
Sekil 3: fig3_eps_inconsistency.png

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Sekil 4: fig4_dist_comparison.png



Sekil 5: fig5_decision_zones.png

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Decision Zones: Should We Continue Optimizing?

